

Introduction to Mathematics

AI Tutor - Mathematics - Lesson 1

What is Mathematics?

Mathematics is the study of numbers, quantities, shapes, and patterns, and the relationships between them. It provides a precise language for describing the world and is the foundation for science, engineering, economics, and technology.

Branches of Mathematics

- **Arithmetic** - the study of numbers and basic operations (addition, subtraction, multiplication, division).
- **Algebra** - using symbols and letters to represent numbers and express relationships.
- **Geometry** - the study of shapes, sizes, angles, and space.
- **Statistics** - collecting, analyzing, and interpreting data.

Why It Matters

Mathematical thinking teaches logical reasoning and problem-solving skills that apply far beyond the classroom - from budgeting money to building software to designing bridges.

Key Terms

- **Variable** - a symbol (often a letter) that represents an unknown value.
- **Equation** - a statement that two expressions are equal.
- **Expression** - a combination of numbers, variables, and operations.